

Insertion Sort - Part 1

In this challenge, you're provided with a nearly sorted list and tasked with inserting the last element into its correct position, ensuring the list remains sorted. This exercise emphasizes understanding the insertion mechanism of the Insertion Sort algorithm.

Example:

Given the array [2, 4, 6, 8, 3], the goal is to insert 3 into its correct position. The process involves shifting elements and results in the following steps:

[2, 4, 6, 8, 8]

[2, 4, 6, 6, 8]

[2, 4, 4, 6, 8]

[2, 3, 4, 6, 8]

Each step involves shifting elements to the right until 3 is placed in its correct position.

Python Code:

```
def insertion_sort1(n, arr):
    key = arr[-1] # The last element to be inserted
    j = n - 2 # Start checking from the second last element

    # Shift elements to the right until we find the correct position for key
    while j >= 0 and arr[j] > key:
        arr[j + 1] = arr[j]
        print(*arr) # Print the array after each shift
        j -= 1

    arr[j + 1] = key # Insert the key at the correct position
    print(*arr) # Print the final sorted array

n = 5
arr = [2, 4, 6, 8, 3]
insertion_sort1(n, arr)
```